

UC LINA 600 LINEAR ALGEBRA FOR GRAPHICS · MODULE 01 · LECTURE NOTES

Why This Is the Graphics Prerequisite

Because the pipeline is four matrix multiplications.

MODULE OUTCOMES

- Trace a vertex through the pipeline at a high level.
- Name the four transforms and what each does.
- Explain why linear algebra is the subject rather than background.
- State what this course assumes and what it supplies.
- Decide whether you can skip it.

1 What happens to a vertex

A VERTEX STARTS as three numbers in the space the artist modelled it in. It ends as a pixel. In between:

- 1 MODEL MATRIX
model space -> world space
"where is this object placed in the scene"
- 2 VIEW MATRIX
world space -> camera space
"what does this look like from the camera"
-- which is a change of basis (Module 04)
- 3 PROJECTION MATRIX
camera space -> clip space
"how does distance become smaller on screen"
-- which needs homogeneous coordinates (M07)
- 4 PERSPECTIVE DIVIDE and VIEWPORT TRANSFORM
clip space -> normalised device coords
-> pixels

S0: $p_{\text{screen}} = \text{Viewport} * \text{Project} * \text{View} * \text{Model} * p$

FOUR MATRIX MULTIPLICATIONS. That is the pipeline, and this whole course is about what each one is.

Four matrix multiplications — that is the pipeline, and the rest of the course is about what each one is and why it has the form it does. Everything in Modules 02 through 08 serves this one line, and

Module 08 returns to it with every matrix derived. It is worth copying this line somewhere and coming back to it as each module fills a piece in.

WHICH IS WHY THIS IS THE SUBJECT RATHER THAN THE BACKGROUND

A graphics course that assumed no linear algebra would have to teach it anyway — there is nothing else in the pipeline — and CSCE 641 spends its time instead on the parts that are not matrices: rasterisation, shading models, texturing, and the hardware.

And the failure mode is specific rather than general: you can copy a projection matrix out of a book and get a picture on the screen — and then be unable to debug it, extend it, or handle the case the book did not cover, such as an off-axis frustum or a reversed depth buffer.

Which happens constantly, because the pipeline is forgiving of ignorance right up until it is not — and the symptom is a scene that is almost right in a way you cannot name: shading slightly off, geometry subtly skewed, something inside out at one camera angle.

So the test is Module 13's and §4's: can you derive the projection matrix rather than recall it? — and deriving it is about four lines once Module 07's homogeneous coordinates are in place, which is the whole argument for spending the four weeks.

2 The four transforms

Matrix	What it actually is	Covered in
Model	Rotation, scale, and translation composed into one matrix — and the order matters.	Modules 03 and 07
View	A change of basis into the camera's frame — not a movement of the camera, but a re-expression of the world.	Module 04
Projection	A shear plus a divide, in disguise — the matrix sets up the divide and the hardware performs it.	Modules 07 and 08
Normal matrix	The inverse transpose of the model-view matrix — and the reason is geometric, not algebraic.	Modules 06 and 09

The view matrix is a change of basis and the projection is a shear — two facts that make the pipeline comprehensible instead of memorised, and both are non-obvious until stated. **The normal matrix row is the one that surprises people:** normals do not transform like points, and **Module 06 §4 explains why** in a way that makes the inverse transpose inevitable rather than arbitrary.

3 Beyond the pipeline

- **CSCE 645 Geometric Modeling** — **splines are linear combinations of basis functions, which is Module 02's vocabulary applied to curves rather than to arrows**, and the whole subject reads differently once you see that.
- **CSCE 647 Image Synthesis** — **every ray-surface intersection is a dot product and a linear solve**, and **importance sampling requires building an orthonormal frame around a normal** (Module 05 §4), which is a Module 05 exercise.
- **CSCE 649 Physically-Based Modeling** — **inertia tensors are eigenvector problems** (the principal axes are the eigenvectors) **and orientations are quaternions** (Modules 10 and 11).
- **CSCE 633 and CSCE 636** — **least squares is the first fitting method you meet** and **principal component analysis is an eigenvector computation** (Modules 10 and 12), so two of this course's later modules are machine learning prerequisites as much as graphics ones.
- **And CSCE 753 Computer Vision, which is this course run backwards** — **graphics computes the image from the transform, and vision recovers the transform from the image** — which is why the camera matrix appears in both and means the same thing.

4 Can you skip it?

FIVE QUESTIONS DECIDE IT, AND THEY ARE THE ONES CSCE 641 ASSUMES IN WEEK TWO

Can you say what a basis is, and explain why the columns of a matrix are where the basis vectors land? (Modules 02 and 03 — and that second fact is the one that makes matrices stop being tables of numbers.)

Can you project one vector onto another, and derive the formula rather than recall it? (Module 05 §2 — and this is the operation every shading model uses.)

Can you say why the cross product's direction depends on a handedness convention, and which convention you use? (Module 06 §3 — and **handedness bugs are the commonest failure in this subject.**)

Can you explain what the fourth coordinate is for (Module 07), and **why normals are transformed by the inverse transpose** (Module 06 §4)? **Five questions; answer all five and skip the course with a clear conscience** — and if you cannot, the ones you missed name the modules that matter most.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
3Blue1Brown — Essence of Linear Algebra (free)	Watch all fifteen before Module 02 — it is three hours and it changes how every later module reads.
Scratchapixel — the geometry and transform sections (free)	§1's pipeline , derived step by step with code, and free.
MIT 18.06, lecture 1 (free video)	§2 — Strang's opening, which frames the whole subject around the column picture.
Gortler, chapters 2 and 3 (library copy)	§§1 and 2 — frames and transforms stated more carefully than most graphics books manage.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Write out the pipeline line** from memory, with all four transforms named.
- 2 **For each transform, say what space it maps from and to.**
- 3 **Find a projection matrix in a book** and list what you do not understand about it.
- 4 **Describe a graphics bug** you have seen that was almost right.
- 5 **Name which module covers each of the four matrices.**
- 6 **Find three places in the program** outside graphics that need this course.
- 7 **Explain why vision is graphics backwards**, in three sentences.
- 8 **Take the five questions** in §4 now, and record your answers.
- 9 **Watch the 3Blue1Brown series** end to end before Module 02.
- 10 **Set up a test scene**: a unit cube, a camera on an axis, and a point whose screen position you can compute by hand.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name the four transforms and the spaces each maps between.
- 2 Write the pipeline as one line.
- 3 Why is linear algebra the subject rather than the background?
- 4 What is the specific failure mode of copying a matrix?
- 5 What is the view matrix, really?

- 6 What is the projection matrix, really?
- 7 Which transform surprises people, and why?
- 8 Name four later courses that need this one, and what each needs.
- 9 Why is computer vision this course backwards?
- 10 Give the five skip-test questions.